1.3 The Smallest Counterexample

WORKSHEET

The question

STEP III 2015, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q5

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Step 4 says "define k to be the smallest positive integer in S". Why are we entitled to do that? Would it be allowed if S were a set of positive *real* numbers?

STAGE 1 Step 3: S is not empty

Covers: Part (i)

Ask yourself if $\sqrt{2} = p/q$, which positive integer, when multiplied by $\sqrt{2}$, gives an integer?

STAGE 2 Step 5: a smaller element

Covers: Part (i)

Ask yourself You need two facts about $n = (\sqrt{2}-1)k$. It is a positive integer, and $n\sqrt{2}$ is an integer. Write $k\sqrt{2} = m$, an integer, and express both n and $n\sqrt{2}$ in terms of k and m.

STAGE 3 Step 6: the contradiction

Covers: Part (i)

Pause and try How big is $\sqrt{2}-1$? Compare $(\sqrt{2}-1)k$ with k, and remember how k was chosen in step 4.

STAGE 4 Cube roots: linking the two numbers

Covers: Part (ii)

Ask yourself Write $a = 2^{1/3}$. If a is rational, is a^2? If a^2 is rational, can you write a in terms of a^2 and $a^3 = 2$?

STAGE 5 Adapting the argument

Covers: Part (ii)

Ask yourself Try to copy part (i), with $\sqrt{2}-1$ replaced by $a-1$. Where does the copy break down?

Pause and try Use a set that remembers more: the positive integers n for which *both* na and na^2 are integers. Check that it is non-empty, and that $(a-1)k$ belongs to it whenever k does.

What's really going on

A proof you can draw

There is a lovely picture behind part (i), due to Stanley Tennenbaum. If $\sqrt{2} = a/b$ then $a^2 = 2b^2$: a square of side a has the same area as two squares of side b. Place the two b-squares in opposite corners of the a-square, as in Figure 1.3.1. They overlap in a small square, and leave two small squares uncovered. Equal areas force the overlap to equal the two uncovered squares together, giving a *smaller* pair of integers with the same property. Smaller and smaller pairs can't go on for ever, so no such squares exist. The smaller pair is exactly $(2b-a,\ a-b)$, which is our $(\sqrt{2}-1)k$ step in disguise.

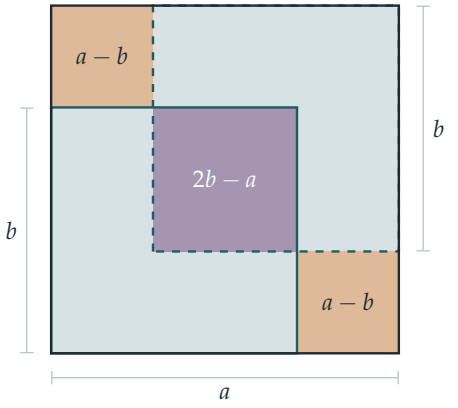


Figure 1.3.1 Tennenbaum's proof. Two squares of side b (teal) sit in opposite corners of a square of side a. If $a^2 = 2b^2$, the area counted twice (the plum overlap) must equal the area missed (the two amber corners), so $(2b - a)^2 = 2(a - b)^2$: a smaller solution.

Well-ordering, induction and descent

The well-ordering principle and the principle of induction are logically equivalent: each can be proved from the other. At university you will see both used as the foundations of the natural numbers. In Section 1.5 we will use the same principle in a different form, Fermat's *method of infinite descent*: there is no infinite strictly decreasing sequence of positive integers.

How far does it go?

The argument proves that $\sqrt{N}$ is irrational whenever N is not a perfect square (see the exercises), and more generally that the nth root of a whole number is either a whole number or irrational. At university this becomes a statement about *algebraic integers*: a rational number that satisfies a monic polynomial equation with integer coefficients, such as $x^3 - 2 = 0$, must be an integer. You will meet the idea again in Section 6.3.

The legend that the Pythagoreans drowned Hippasus for revealing the irrationality of $\sqrt{2}$ is probably untrue. What is true is that the discovery forced Greek mathematicians to separate the study of number from the study of length, a split that took until the nineteenth century, and Dedekind's construction of the real numbers, to heal.

Try this next

1 **Other square roots.** Adapt the argument to show that $\sqrt{3}$ is irrational. Which number plays the role of $\sqrt{2} - 1$?

2 **All non-squares.** Let N be a positive integer that is not a perfect square, and let m be the integer with $m < \sqrt{N} < m + 1$. Prove that $\sqrt{N}$ is irrational by considering

$(\sqrt{N} - m)k$.

3 **Spot the flaw.** Try to run the argument for $\sqrt{4}$. Exactly which step fails?

4 **Tennenbaum's picture.** Use Figure 1.3.1 to show that if $a^2 = 2b^2$ then $(2b - a)^2 = 2(a - b)^2$. Check this identity algebraically too.

Further reading

- **Daniel Velleman, *How to Prove It*** (Cambridge University Press). A thorough introduction to proof, including induction and well-ordering.
- **R. P. Burn, *A Pathway into Number Theory*** (Cambridge University Press). A book made entirely of carefully sequenced questions; ideal preparation for a first-year course.
- **John Conway & Joseph Shipman, "Extreme proofs I: the irrationality of $\sqrt{2}$"** (*Mathematical Intelligencer*, 2013). Many proofs of one fact, compared with wit.

The full guided solution is in Section 1.3:
https://beyondstep.keithmcnulty.org/ch01/well-ordering.html